LESSON 1.1
REWRITING RATIONAL NUMBERS IN EQUIVALENT FORMS

LESSON INTRODUCTION

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

LEARNING TARGETS

- I can rewrite rational numbers in different but equivalent forms.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.

WORKING TOWARD

8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

ESSENTIAL QUESTIONS

- What are rational numbers?
- How can you convert between fractions, decimals, and percentages?
- How can you convert a repeating decimal to a fraction?
- What is the difference between a terminating and a repeating decimal?

LESSON NARRATIVE

In this lesson, students will discover how to rewrite rational numbers in different but equivalent formats. Students will convert between fractions, decimals, and percentages. Students will also determine if a fraction turns into a repeating or terminating decimal. Students will learn how to use substitution to convert a repeating decimal into a fraction.

COMPONENT SUMMARY

1.1.1 WARM-UP *(~5 MIN)*

Discuss a video about working as a designer

1.1.2 ACTIVITY *(10-15 MIN)*

Match fraction with its equivalent decimal and percent

1.1.3 GUIDED PRACTICE *(10-15 MIN)*

Determine if a decimal is terminating or repeating practice

1.1.4 GUIDED INSTRUCTION *(10-15 MIN)*

Introduction to using an equation to convert repeating decimal into fraction

1.1.5 TRY IT! *(10-15 MIN)*

Independently determine equivalent forms (fraction, decimal, or percent) of a number

1.1.6 WRAP-UP *(~5 MIN)*

Reflect on understanding of lesson

RESOURCES

GLOSSARY

Key Terms:

- Rational number
- Repeating decimal

Additional Terms:

- N/A

MATERIALS

- Cards for sort activity
- Calculator
- (Optional) paper

ADDITIONAL PREPARATION

- 1.1.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may divide by 100 instead of multiplying when converting from a decimal to a percent.
- Students may put 100 in the denominator of a repeating decimal to make it a fraction.
- Students may leave a numerator as a decimal.

THINGS TO CONSIDER

- Consider allowing students to use a calculator throughout the lesson as the division is not the focus but rather the process that is occurring.
- Consider discussing with students why a decimal cannot be in the numerator of a fraction, and why a repeating decimal cannot have a denominator of 100.

LITERACY AND LANGUAGE ROUTINES

Take Turns

Use this activity to determine where students are at with rational numbers. If in groups/pairs, have students take turns matching so each student has the opportunity to match the equivalent forms.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

1.1.4 Guided Instruction provides students with an opportunity to reflect on and discuss how to determine which and how many digits need to be subtracted from a repeating decimal in order to eliminate the repeating part. Select, sequence, and present student work to advance and deepen understanding of correct and increasingly efficient methods.

DIFFERENTIATE INSTRUCTION

Consider turning closed caption on as students are watching the video for those who are hearing-impaired. After students have completed this activity, consider having them stand up for terminating or sit down for repeating, and poll the class for their answers.

Additional differentiation opportunities are denoted throughout the lesson guide.

1.1.1 WARM-UP: DAY AT WORK: SENIOR DESIGNER

TEACHER MOVES

Consider allowing students to watch the corresponding video independently if technology is available and have them take notes as they are watching.

TEACHER MOVES

Consider posting the definition of the word “cultivating” and providing an example to assist students with their responses.

DIFFERENTIATE INSTRUCTION

Consider turning closed caption on as students are watching the video for those who are hearing-impaired.

1.1.1 Warm-Up: Day at Work: Senior Designer

Nik Brensnick is a senior product designer. She is quoted as saying, “A big part of working and being successful in the workplace has to do with cultivating relationships with other people.”

1. Why do you think Nik has this belief about being successful in the workplace?
   Answers will vary. Nik says this because building relationships with others gives us a sense of belonging. She also mentioned that “if you are fun to work with then you will always have work.”

1.1.2 Activity: Match It Up!

1. Work with your partner to match the equivalent rational numbers expressed in different forms. Record each matching set in the table.

Fraction	Decimal	Percent
$\frac{2}{3}$	$0.\overline{6}$	$66.\overline{6}\%$
$\frac{3}{5}$	0.6	60%
$\frac{5}{12}$	$0.41\overline{6}$	$41.\overline{6}\%$
$\frac{17}{20}$	0.85	85%
$\frac{13}{10}$	1.3	130%
$\frac{7}{8}$	0.875	87.5%
$\frac{1}{4}$	0.25	25%
$\frac{22}{25}$	0.88	88%
$\frac{6}{8}$	0.75	75%
$\frac{2}{9}$	$0.\overline{2}$	$22.\overline{2}\%$

2. Explain your method for matching the equivalent rational forms.
Answers will vary. When matching the equivalent rational forms, I started with the fractions and then moved to matching the decimal form cards using the calculator.

1.1.2 ACTIVITY: MATCH IT UP!

TEACHER MOVES

Take Turns

Use this activity to determine where students are at with rational numbers. If in groups/pairs, have students take turns matching so each student has the opportunity to match the equivalent forms.

DIFFERENTIATE INSTRUCTION

Consider making this into a digital matching activity, using Google Slides or PowerPoint, to drag and drop the numbers into the correct column in the table.

1.1.3 GUIDED PRACTICE: REPEATING DECIMALS

DIFFERENTIATE INSTRUCTION

After students have completed this activity, consider having them stand up for terminating or sit down for repeating, and poll the class for their answers.

TEACHER MOVES

Students should be allowed to use a calculator for this section to allow them to focus on the number patterns, as solving is not the focus of this activity.

ENRICHMENT

Why do the digits repeat? Is there a limit to how many digits repeat? After how many digits could you decide that the decimal probably does not repeat? If only the last number repeats, how should you write the number with the bar?

1.1.3 Guided Practice: Repeating Decimals

A ***rational number*** is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.

1. Consider the fractions in the table shown.

   a. Complete the table by rewriting each fraction into its equivalent form. If a decimal repeats digits in a pattern, use an ellipse (…) to indicate that the pattern continues.

Fraction Form	Decimal Form	Terminating or Repeating?
$\frac{1}{6}$	$0.166\ldots$	○ terminating ○ repeating
$\frac{5}{8}$	0.625	○ terminating ○ repeating
$\frac{43}{9}$	$4.77\ldots$	○ terminating ○ repeating

   b. Based on your work in Part A, explain whether it is possible for a repeating decimal to ever terminate.
   Answers will vary. It is not possible for a repeating decimal to ever terminate. Repeating decimals continue infinitely.

A number with a decimal that never ends, but has a pattern that repeats infinitely, is called a **repeating decimal.**

Repeating decimals are written with a bar over the digits that repeat.

For example, the fraction $\frac{17}{6}$ is expressed as $2.8\overline{3}$ when written as a decimal because the 3 is the digit that repeats infinitely.

2. Dulce divided 73 by 99 using a calculator to rewrite $\frac{73}{99}$ in decimal form. She found the resulting pattern $0.737373\ldots$

   Write $\frac{73}{99}$ in its equivalent decimal form.

   $0.\overline{73}$

1.1.4 Guided Instruction: Rewriting Repeating Decimals as Fractions

1. Consider the repeating decimal $0.4\overline{1}$.

   a. If $x = 0.4\overline{1}$, complete each equation.

   $$x = 0.4\overline{1}$$
   $$10x = \underline{4.\overline{1}}$$
   $$100x = \underline{41.\overline{1}}$$

   b. Which two values, when subtracted, would result in the elimination of the repeating portion of the decimal?

   $4.1\overline{1} - 0.4\overline{1} = 3.7$

   c. Use two of the relationships represented in the equations in Part A to complete the steps to find the fraction equivalent of $0.4\overline{1}$.

   $$\underline{10x} - \underline{x} = \underline{4.1\overline{1}} - \underline{0.4\overline{1}}$$
   $$\underline{9}\,x = \underline{3.7}$$
   $$x = \frac{\boxed{3.7}}{\boxed{9}}$$

   d. Rewrite the solution as a fraction in the simplest form, if possible.

   $\frac{37}{90}$

1.1.4 GUIDED INSTRUCTION: REWRITING REPEATING DECIMALS AS FRACTIONS

TEACHER MOVES

MA.K12.MTR.4.1: Discussion and Reflection

Have students discuss the following questions with a partner:

- Why should 4.11 be chosen to subtract 0.41 from?
- Why is it necessary to eliminate the repeating decimal? How can you use substitution to solve the problem?
- Which numbers and variables can be substituted for each other?

ENRICHMENT

Why do the numerator and denominator both increase when simplifying this fraction? Would the decimal still repeat if the fraction was converted back to a decimal?

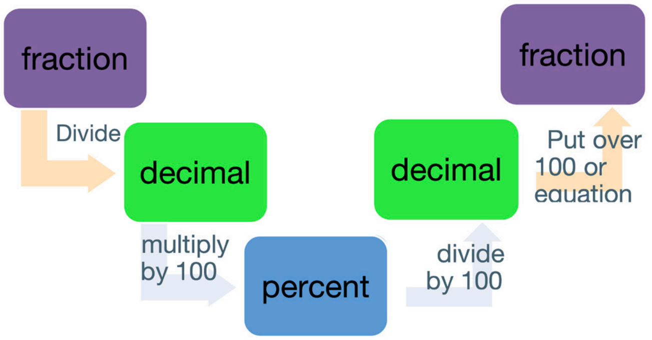
1.1.5 TRY IT: EQUIVALENT FORMS

DIFFERENTIATE INSTRUCTION

Consider providing the following visual for students to use when converting:

TEACHER MOVES

Encourage students to notice that even though we read from left to right, the conversions do not need to always occur in the order of the table.



1.1.5 Try It: Equivalent Forms

1. Complete the table.

Fraction	Decimal	Percent
$\frac{15}{99}$	$0.\overline{15}$	$15.\overline{15}\%$
$\frac{8}{25}$	0.32	32%
$\frac{8}{11}$	$0.\overline{72}$	$72.\overline{72}\%$
$\frac{5}{16}$	0.3125	31.25%
$\frac{8}{9}$	$0.\overline{8}$	$88.\overline{8}\%$
$\frac{47}{90}$	$0.5\overline{2}$	$52.\overline{2}\%$
$\frac{1}{12}$	$0.08\overline{3}$	$8.\overline{3}\%$

1.1.6 Wrap-Up: Reflection

1. Check all the statements below that apply to your understanding.

   - ☐ I understand how to rewrite rational numbers in different but equivalent forms.
   - ☐ I need support on how to rewrite rational numbers in different but equivalent forms.
   - ☐ I understand how to convert a repeating decimal to a fraction.
   - ☐ I need support on how to convert a repeating decimal to a fraction.

   Answers will vary.

1.1.6 WRAP-UP: REFLECTION

TEACHER MOVES

Before students complete the Wrap-Up, consider allowing them to write everything they learned from the lesson and then match what they wrote to the options in the reflection.

DIFFERENTIATE INSTRUCTION

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.